THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH2241 Introduction to Mathematical Analysis
Tutorial 4

1. Prove or disprove: $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^{1+\frac{1}{n}}}$ is a divergent series.
   (*Hint: Note that* $2^n > n$ *for any* $n \in \mathbb{N}$)

2. **(Limit Comparison Test)**
   Let $\{a_n\}$ and $\{b_n\}$ be two sequences of positive terms and $\lim_{n\to\infty} \frac{a_n}{b_n} = L$ for some $L \geq 0$.

   (a) Prove that if $L > 0$, then $\displaystyle\sum_{n=1}^{\infty} a_n$ converges $\iff$ $\displaystyle\sum_{n=1}^{\infty} b_n$ converges.

   (b) Prove that if $L = 0$ and if $\displaystyle\sum_{n=1}^{\infty} b_n$ converges, then $\displaystyle\sum_{n=1}^{\infty} a_n$ converges.

   (c) Give an example of sequences $\{a_n\}$ and $\{b_n\}$ of positive terms such that $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$, $\displaystyle\sum_{n=1}^{\infty} a_n$ converges but $\displaystyle\sum_{n=1}^{\infty} b_n$ diverges.

   (d) Determine whether or not the series $\displaystyle\sum_{n=1}^{\infty} \frac{n^2+n-1}{27n^5-n^4+9n^3-2n+6}$ converges.

3. Prove that the series $\dfrac{1^2}{0!} + \dfrac{2^2}{1!} + \dfrac{3^2}{2!} + \dfrac{4^2}{3!} + \cdots$ converges and find its sum.
   (*Hint: You may use the fact that* $\sum_{n=0}^{\infty} \frac{1}{n!} = e$.)

4. Prove each of the following statements:

   (a) If $\displaystyle\sum_{n=1}^{\infty} a_n$ converges, then $\displaystyle\sum_{n=1}^{\infty} \cos a_n$ diverges.

   (b) If $\displaystyle\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\displaystyle\sum_{n=1}^{\infty} \sin a_n$ converges absolutely.

5. Determine whether each of the following series converges absolutely, converges, or diverges.

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n+\sqrt{n}}$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1} n}{n+\sqrt{n}}$

6. (a) **(Cauchy Condensation Test)**
Let $\{a_n\}$ be a decreasing sequence with $a_n \geq 0$ for any $n \in \mathbb{N}$. Prove that $\displaystyle\sum_{n=1}^{\infty} a_n$ converges $\iff \displaystyle\sum_{n=0}^{\infty} 2^n a_{2^n}$ converges.

(b) Let $p > 0$. Use the Cauchy Condensation Test to prove that the p-series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$.